



HK IMO Training 2019-2020
Problem Set 6 Suggested Solutions



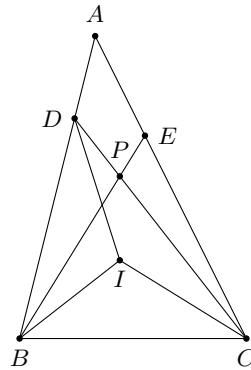
Problem 21. Let ABC be an acute triangle, where $AB < AC$, and let M be the midpoint of BC , and K be the midpoint of the polygonal line BAC . Show that $\sqrt{2}KM > AB$.

Solution. Let $BC = a$, $CA = b$ and $AB = c$. Then

$$\begin{aligned} KM^2 &= KC^2 + MC^2 - 2KC \times MC \cos \angle KCM \\ &= \left(\frac{b+c}{2}\right)^2 + \left(\frac{a}{2}\right)^2 - 2\left(\frac{b+c}{2}\right)\left(\frac{a}{2}\right) \cos \angle ACB \\ &= \frac{(b+c)^2}{4} + \frac{a^2}{4} - \frac{a(b+c)}{2} \cdot \frac{a^2 + b^2 - c^2}{2ab} \\ &= \frac{1}{4b}(b(b+c)^2 + a^2b - (b+c)(a^2 + b^2 - c^2)) \\ &= \frac{c(b^2 + c^2 + 2bc - a^2)}{4b}. \end{aligned}$$

We need to prove $2KM^2 > c^2$, i.e. $b^2 + c^2 + 2bc - a^2 > 2bc$. This is true since $b^2 + c^2 - a^2 > 0$ from the fact that ABC is acute.

Problem 22. The points D and E are chosen on the sides AB and AC of a triangle ABC so that $DB = BC = CE$. The segments BE and CD meet at P . Show that the circumcircles of the triangles BDP and CEP meet at the incentre of $\triangle ABC$.



Solution. Let I be the incentre of $\triangle ABC$. Since $BC = BD$, we have $\angle BDC = \angle BCD = \frac{180^\circ - B}{2}$. Similarly, we have $\angle CBE = \angle CEB = \frac{180^\circ - C}{2}$. Then

$$\begin{aligned} \angle BPD &= 180^\circ - \angle BDP - \angle DBP \\ &= 180^\circ - \frac{180^\circ - B}{2} - \left(B - \frac{180^\circ - C}{2}\right) \\ &= 180^\circ - \frac{B+C}{2} \\ &= 90^\circ + \frac{A}{2} \\ &= \angle BIC. \end{aligned}$$

Since BI is the angle bisector of the isosceles triangle BCD , we have $\angle BIC = \angle BID$. Therefore, $\angle BPD = \angle BID$, hence B, D, P, I are concyclic. By symmetry, I also lies on the circumcircle of $\triangle CEP$ as desired.

Problem 23. In a class, there are m students. During September, each of them visited the pool several times. No one went to the pool twice in a day. It turned out that during September, all students visited the pool different number of times. Moreover, for any two students A and B there was a day when A visited the pool, while B did not, and vice versa. Determine the largest possible value of m .

Solution. There are 30 days in September. If we use a set to include all the days that a certain student visited the pool, we simply need to find the maximal number of subsets of $\{1, 2, \dots, 30\}$ having different cardinalities such that none is a subset of another.

We claim that the answer is 28. An example of the 28 subsets is

$$\begin{aligned} &\{29, 30\}, \{27, 28, 30\}, \{25, 26, 28, 30\}, \dots, \{3, 4, 6, 8, 10, \dots, 28, 30\}, \\ &\quad \{1, 2, 3, 6, 8, 10, \dots, 28, 30\}, \\ &\quad \{1, 2, 3, 4, 5, 8, 10, \dots, 28, 30\}, \\ &\quad \dots, \\ &\quad \{1, 2, 3, 4, 5, 6, 7, \dots, 26, 27, 30\}, \\ &\quad \{1, 2, 3, 4, 5, 6, 7, \dots, 26, 27, 28, 29\}. \end{aligned}$$

It is easy to see that each subset has one more element than the previous one, so they have different cardinalities. Note that the smallest subset $\{29, 30\}$ is not a subset of the others. Also, the largest subset $\{1, 2, \dots, 29\}$ does not contain any other set since all other sets contain 30. By removing these two subsets, and removing the common element 30 from the remaining subsets, we are back to the analogous construction with 30 replaced by 28. Then we argue inductively and hence the sets are not subsets of each other.

Next, we prove that the number of subsets cannot exceed 28. If there are at least 29 subsets, as they are nonempty and cannot be the entire set $\{1, 2, \dots, 30\}$ (otherwise it contains any other subset), their cardinalities must be $1, 2, \dots, 29$ respectively. WLOG assume the singleton set is $\{30\}$. Then the subset with 29 elements must be $\{1, 2, \dots, 29\}$. But then any other subset must contain $\{30\}$ or is a subset of $\{1, 2, \dots, 29\}$. This is a contradiction. Therefore, the answer is 28.

Problem 24. Prove that for each positive integer n there are at most two pairs (a, b) of positive integers with following two properties:

- (i) $a^2 + b = n$,
- (ii) $a + b$ is a power of two, i.e. there is an integer $k \geq 0$ such that $a + b = 2^k$.

Solution. Suppose there are three or more of such pairs for one particular n . Then by the pigeonhole principle, there are at least two such pairs, say (a, b) and (c, d) , with $a \equiv c \pmod{2}$. Write $a + b = 2^k$ and $c + d = 2^\ell$ where k and ℓ are positive integers. Without loss of generality, assume that $\ell \geq k$. We see that

$$\begin{aligned} 2^\ell - 2^k &= (c + d) - (a + b) \\ &= (c + n - c^2) - (a + n - a^2) \\ &= c - a - c^2 + a^2 \\ &= (a + c - 1)(a - c). \end{aligned}$$

As $\ell \geq k$, we have that 2^k is a divisor of $2^\ell - 2^k$. Because $a \equiv c \pmod{2}$, the integer $a + c - 1$ is odd, hence 2^k is a divisor of $a - c$. As $2^\ell - 2^k \geq 0$ and $a + c - 1 > 0$, we have $a - c \geq 0$. On the other hand, we have $a - c < a + b = 2^k$, hence $0 \leq a - c < 2^k$, while $2^k \mid a - c$. We conclude that $a - c = 0$, hence $a = c$. Now we have $b = n - a^2 = n - c^2 = d$, hence the two pairs (a, b) and (c, d) are identical, which gives a contradiction. Therefore, there are at most two pairs (a, b) having the two properties.